Sample Work

Grade-Wise AOD Addition Pattern and Nitrogen-to-Argon Switch-over Standardisation

Role

Operations Analyst

Company

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Department

Steel Melting Shop, AOD Operations

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1. Project Overview

This sample work is based on the standardisation of grade-wise addition patterns in the Argon Oxygen Decarburisation process used in stainless steel production.

The work focused on creating a structured operating reference for 300 series, 400 series, and 200 series stainless steel grades. Each grade series required a different addition pattern, flux quantity, alloy sequence, temperature control, stirring phase, reduction condition, and nitrogen-to-argon switch-over logic.

The objective was to improve process consistency, reduce dependency on individual operator experience, support productivity improvement, and ensure better control during AOD operations.

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2. Business Problem

The AOD process involves multiple grade families and each grade has specific process requirements. If addition sequence, flux quantity, switch-over timing, or temperature range is not controlled properly, it can affect process time, chemical accuracy, productivity, and operational consistency.

Key problems identified were:

Area	***Problem***
Grade-wise variation	300, 400, and 200 series required different addition patterns
Process dependency	Operators depended heavily on experience and verbal instruction
Flux control	Flux quantity differed by grade series and needed standard control

Gas switch-over	Nitrogen-to-Argon switch-over timing differed by grade
Temperature control	Sampling and reduction temperatures needed defined ranges
Quality control	Incorrect timing could affect chemistry achievement
Shift consistency	Different shifts needed the same standard reference
Safety	Dog house closure and shop-floor safety needed clear instruction

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3. Objective

The objective was to create a standard grade-wise operating reference for AOD operations.

The main objectives were:

Objective	**Description**
Standardise addition pattern	Define the correct material addition sequence for each grade series
Improve process consistency	Ensure all shifts follow the same process flow
Improve productivity	Reduce process variation and avoid unnecessary delay
Improve chemistry control	Support better control of carbon, chromium, nickel, manganese, nitrogen, sulphur, and other elements
Improve gas control	Define Nitrogen-to-Argon switch-over timing by grade
Improve safety compliance	Document key safety points for shop-floor execution
Support training	Help new or less experienced team members understand grade-wise AOD control

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4. Scope

The scope of this work covered grade-wise AOD addition pattern and gas switch-over control.

The scope included:

Area	***Included in Scope***
300 series	Addition pattern for 300 series grades

400 series	Addition pattern for 400 series grades
200 series	Addition pattern for 200 series grades
Grade-specific controls	304L, 310S, 409L, 439, 441-Honda, and 210LN controls
Flux addition	Main blow, DB1, and DB2 flux addition
Alloy addition	HCFeCr, HCFeNi, DRI, LCFeNi, HCFeMn, and other additions
Temperature control	Sampling and reduction temperature ranges
Gas switch-over	Nitrogen-to-Argon switch-over timing and Argon consumption reference
Safety	Dog house closure and operational precautions

5. Stakeholders

STAKEHOLDER	RESPONSIBILITY
SHIFT-IN-CHARGE	Control grade-wise execution during AOD operation
AOD OPERATOR	Follow the defined addition sequence and process instructions
PRODUCTION MANAGER	Monitor productivity, heat performance, and shift compliance
QUALITY TEAM	Verify chemistry achievement and grade-specific quality requirements
SAFETY TEAM	Ensure safe operation and shop-floor compliance
MAINTENANCE TEAM	Ensure equipment, ladle, launder, chute, and porous plug readiness
GAS SUPPLIER / BOC COORDINATION	Support Argon backup pump cooling requirement for specific grades

6. As-Is Process

Before standardisation, grade-wise AOD process control was largely based on operator experience, shift knowledge, and practical understanding of different stainless steel grades.

Area	As-Is Situation
Grade-wise process	Grade-specific instructions were followed through experience
Addition sequence	Process flow was not always available in one structured reference

Gas switch-over	Switch-over timing varied by grade and required clear documentation
Flux quantity	Different series required different flux quantities
Safety instruction	Safety points needed to be clearly visible in the operating reference
Training	New team members needed senior guidance for grade-specific practices

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7. To-Be Process

The To-Be process created a structured reference for grade-wise AOD addition patterns and switch-over rules.

Area	To-Be Process
Process flow	Standard process flow created for 300, 400, and 200 series
Addition control	Material addition sequence defined for each grade series
Flux control	Flux quantity and addition stages clearly defined
Gas switch-over	Nitrogen-to-Argon switch-over timing documented grade-wise
Temperature control	Sampling and reduction temperature ranges defined
Safety control	Safety points included in the operating reference
Shift consistency	Same reference available for all shifts

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8. Standard Addition Pattern for 300 Series

The 300 series addition pattern was structured to control flux, alloy addition, main blow, sampling, stirring, and reduction.

Step	Activity
1	Add zero batch with 4 MT + 1 MT flux
2	Start pouring
3	Start process
4	Add 3 MT Dololime
5	Add HCFeCr
6	Add 1 to 1.5 MT flux

7	Add HCFeNi
8	Add DRI / LCFeNi
9	Add HCFeMn
10	Add flux up to 12 MT
11	Continue main blow
12	Add remaining flux in DB1 and DB2
13	Take sample and temperature in the range of 1700 to 1720°C
14	Complete stirring phase for minimum 2 minutes
15	Complete reduction in the range of 1690 to 1710°C

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9. Standard Addition Pattern for 400 Series

The 400 series addition pattern required a separate flow because the alloy addition and reduction requirements differed from the 300 series.

Step	Activity
1	Add zero batch with 4 MT + 1 MT flux
2	Start pouring
3	Start process
4	Add 3 MT Dololime
5	Add HCFeCr
6	Add 1 MT flux
7	Add DRI
8	Add HCFeMn
9	Add flux up to 12 MT
10	Continue main blow
11	Add remaining flux in DB1 and DB2
12	Take sample and temperature in the range of 1700 to 1720°C
13	Complete stirring phase for minimum 2 minutes

14	Complete reduction in the range of 1690 to 1710°C for minimum 6 minutes

10. Standard Addition Pattern for 200 Series

The 200 series addition pattern required higher flux addition compared with 300 and 400 series.

Step	Activity
1	Add zero batch with 4 MT + 1 MT flux
2	Start pouring
3	Start process
4	Add 3 MT Dololime
5	Add HCFeCr
6	Add 4 MT flux
7	Add DRI
8	Add HCFeMn
9	Add flux up to 14 MT
10	Continue main blow
11	Add remaining flux in DB1 and DB2
12	Take sample and temperature in the range of 1700 to 1720°C
13	Complete stirring phase for minimum 2 minutes
14	Complete reduction in the range of 1690 to 1710°C for minimum 6 minutes

11. Grade-Specific Process Controls

Different stainless steel grades required specific operating controls. These controls were documented to reduce process variation and improve grade-wise execution.

Grade / Series	Key Control Points
304L / EN1.4307	HCFeCr to be added slowly before the defined oxygen requirement. Flux should not be added while HCFe alloy addition is ongoing. Sampling temperature should be around 1720°C. Tapping temperature should be under 1700°C.

310S	Argon backup pump cooling to be started before heat. Nitrogen PPM should be less than 150. Extra flux should not be used. Ladle life, launder condition, slide gate plate, purging, and slag control should be monitored.
409L / 439 / 441-Honda	Argon backup pump cooling to be arranged before heat. MS bundle and LPFeCr availability should be ensured. Dome cleaning and mouth cleaning should be completed before heat. Both porous plugs should be working.
210LN	Low build-up, low tapping sulphur, correct opening phosphorus range, report checking, and Mn metal availability should be ensured. Excess flux should not be added.

12. Nitrogen-to-Argon Switch-over Control

Nitrogen-to-Argon switch-over timing was defined grade-wise to control nitrogen levels and improve process stability.

Grade Family	Switch-over Control
300 series	Switch-over timing varied by grade. Some grades required switch-over before VCD, during reduction, or in second blow.
400 series	Switch-over timing was defined during main blow or reduction depending on grade requirement.
200 series	Switch-over timing was mainly linked to percentage of reduction stage.

Examples of switch-over logic:

Grade	Nitrogen Range	Switch-over Condition
304	275 to 475 ppm	2 minutes before VCD
304L	850 to 975 ppm	38% of reduction
310S	150 ppm max	Second blow
316L	250 to 500 ppm	On reduction
409L	80 ppm max	30% of main blow
410DB	Not specified	50% of main blow
441-Honda	125 ppm max	30% of main blow
201LN	1600 to 1800 ppm	85% of reduction

201	1300 to 1400 ppm	56% of reduction
J4	1200 to 1400 ppm	78% of reduction

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13. Business Rules

Rule ID	Business Rule
BR-01	Each stainless steel series must follow its own defined AOD addition pattern.
BR-02	Zero batch must be added before pouring.
BR-03	Dololime must be added after process start as per standard quantity.
BR-04	Flux addition must follow grade-wise quantity limits.
BR-05	300 series flux must be added up to 12 MT during main blow.
BR-06	400 series flux must be added up to 12 MT during main blow.
BR-07	200 series flux must be added up to 14 MT during main blow.
BR-08	Remaining flux must be added during DB1 and DB2.
BR-09	Sampling temperature should be maintained between 1700 and 1720°C.
BR-10	Reduction temperature should be maintained between 1690 and 1710°C.
BR-11	Stirring phase should be minimum 2 minutes.
BR-12	For 400 and 200 series, reduction should be minimum 6 minutes where applicable.
BR-13	Nitrogen-to-Argon switch-over must be followed as per grade-specific requirement.
BR-14	Dog house must remain closed until tapping is complete.
BR-15	Excess flux should not be added for grades where it is restricted.

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14. Safety Requirements

Safety requirements were included as part of the operating reference.

Safety Area	Control
Dog house	Must be closed until tapping is complete
Ladle condition	Ladle and launder must be clean before critical grades

Porous plug	Both porous plugs should be working where required
Chute condition	Reduction mixture chute must be clean
Purging	Purging should be enough to show bubbles on the surface
Mouth and dome	Dome cleaning and mouth cleaning should be completed before heat where required

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15. Process Controls

Control Area	Control Description
Grade-wise process flow	Separate addition pattern for 300, 400, and 200 series
Flux control	Flux quantity defined by series and process stage
Alloy addition	Sequence defined for HCFeCr, HCFeNi, DRI, LCFeNi, and HCFeMn
Temperature control	Sampling and reduction temperatures defined
Stirring control	Minimum stirring time defined
Reduction control	Reduction temperature and minimum duration defined
Gas switch-over	Nitrogen-to-Argon switch-over timing defined by grade
Safety control	Dog house closure and equipment readiness included

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16. Operations Analyst View

This work involved analysing grade-wise production requirements and converting them into a structured operational reference.

Operations analysis activities included:

Activity	Description
Process review	Reviewed AOD addition patterns for 300, 400, and 200 series
Process comparison	Compared differences in flux quantity, alloy sequence, and reduction control
Standardisation	Created a structured reference for grade-wise operations
Control mapping	Mapped temperature, flux, gas switch-over, and safety controls
Risk reduction	Reduced chances of incorrect sequence or wrong switch-over timing

Productivity focus	Supported process consistency and reduced operational variation

17. Expected Benefits

Area	Expected Benefit
Process consistency	Same grade-wise method followed across shifts
Productivity	Reduced process variation and better heat control
Chemistry control	Better control of grade-specific element ranges
Gas efficiency	Improved Nitrogen-to-Argon switch-over discipline
Training	Easier reference for new and less experienced team members
Safety	Clear safety points included in the process
Shift compliance	Shift-In-Charges can follow one standard reference
Quality support	Better control over grade-wise production requirements

18. KPI Tracking

KPI	Before Standardisation	After Standardisation
Grade-wise process consistency	Dependent on operator experience	Based on documented process flow
Flux addition control	Variable by shift practice	Controlled by grade series
Gas switch-over timing	Experience-based	Defined by grade requirement
Sampling temperature	Monitored during process	Standard range clearly documented
Reduction control	Grade-dependent knowledge	Documented temperature and time control
Safety compliance	Verbal instruction and practice	Safety points clearly captured
Training dependency	High dependency on senior operators	Reduced through structured reference

19. Final Summary

The Grade-Wise AOD Addition Pattern and Nitrogen-to-Argon Switch-over Standardisation work focused on improving process consistency, grade-wise control, and shift-level execution in AOD operations.

The work involved analysing the addition patterns for 300, 400, and 200 series stainless steel grades, documenting alloy and flux addition sequences, defining sampling and reduction temperature ranges, capturing grade-specific process controls, and standardising Nitrogen-to-Argon switch-over timing.

This sample demonstrates practical experience in operations analysis, process mapping, business rules documentation, grade-wise process control, stakeholder coordination, risk reduction, and productivity-focused process improvement in a manufacturing environment.